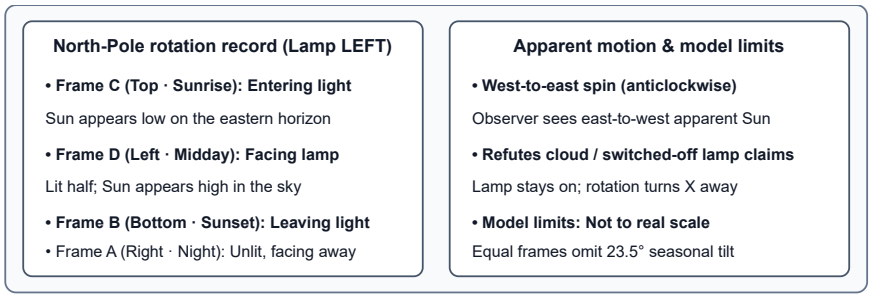
Reference sheets

Reference A · Four-frame North-Pole rotation record (Fixed lamp on LEFT)

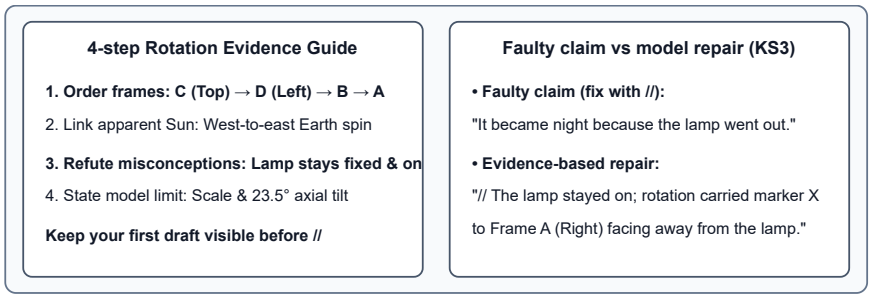


North-Pole rotation record. It links Frames C, D, B, and A to apparent Sun motion and limits.

Reference only. Do not write on this sheet. Use these supplied observations to support your rotation explanation.

Ordered frame & position.	Illumination state of marker X.	Apparent Sun position from marker X.	Rotation clue & model note.
1. Frame C · Top (Sunrise).	Entering the lit left half from shadow.	Sun appears low on the eastern horizon.	Start of cycle (0°); Earth spins anticlockwise (west to east).
2. Frame D · Left (Middle of day).	Facing the fixed lamp directly in the lit half.	Sun appears high across the sky.	After 1st quarter turn (90°); lamp stays fixed and on.
3. Frame B · Bottom (Sunset).	Leaving the lit left half into shadow.	Sun appears low on the western horizon.	After 2nd quarter turn (180°); observer turns away from lamp.
4. Frame A · Right (Night).	Facing away from the lamp in the unlit right half.	Sun is below the local horizon (darkness).	After 3rd quarter turn (270°); 4th turn returns X to Top (360°).

Reference B · Refuting misconceptions, apparent motion, and model limits



Four-step Rotation Evidence guide. It shows frame ordering, apparent motion, refuting cloud claims, and limits.

Evidence step.	Sentence stem for your book.	Key Stage 3 check.
1. Order the 24-hour cycle.	From sunrise, marker X moves through C (Top), D (Left), B (Bottom), and A (Right).	Did you track the same marker back to its starting point?
2. Explain apparent Sun motion.	Earth rotates west to east, so the Sun appears to move east to west.	Did you distinguish observer viewpoint from real solar motion?
3. Refute 'clouds/lamp off' claims.	No clouds are present and the lamp stays on. Rotation turns X into shadow.	Did you use Frame D and Frame A to prove orientation changed?
4. State one honest model limit.	// Limit: size and distance are not to scale. Equal frames omit 23.5° tilt.	Did you state what equal quarter turns cannot show?

Made by Matt

Tier pack · Independent

Use Reference A and Reference B. Work in your book or on this sheet. Point, speak, sign, draw, type, write, or tell an adult. You may pass.

Arrival · 4 questions

1. Use Box 1 of the arrival drawing. Distinguish Earth's daily rotation from Earth's yearly orbit.

Help. Contrast spinning on an axis with travelling around the Sun.

2. Look at Box 1 and Box 2. Marker X turns from the unlit half into the lit half. Which daily moment is this?

Help. Sunrise enters light; sunset leaves light.

3. Look at Box 2. The lamp stayed on while marker X became unlit. What physical cause does this observation support?

Help. State what moved while the lamp stayed fixed.

4. Look at Box 3. How do we verify that we tracked the same place through one complete cycle?

Help. Track one marker back to its starting position.

You do tasks

1. Use Reference A. Complete a four-row evidence record for marker X (Frames C, D, B, and A). State the position, illumination state, and apparent Sun position. **Staff I ☐ S ☐ C ☐**

Help. Include Top, Left, Bottom, and Right and how the Sun appears from X.

2. Use Reference A. Explain why the Sun appears to move east to west across our sky. State what moves and what stays fixed in the model. **Staff I ☐ S ☐ C ☐**

Help. Link west-to-east Earth rotation to the observer's viewpoint.

3. Use Reference B. Use model evidence to refute 'Clouds cause night.' State one model limit, and add one // revision after feedback. **Staff I ☐ S ☐ C ☐**

Help. Check Steps 3 and 4 on Reference B.

Optional participation-loop prompt. Share your four-frame evidence and limit with an adult. Hear one feedback step. Reply below or with // in your book. (Staff: the pupil responds to feedback with an adult, and R is marked only after the pupil has responded.)

Exit ticket

1. Use Frame D (Left) and Frame A (Right) to explain how day changes into night. State what moved and what stayed fixed.

2. Why does the Sun appear to move east to west each day?

3. State one scientific limit of the four-frame classroom globe and lamp model.

Pupil voice

What helped you today, and what would you change?

Point, speak, sign, draw, write or choose. You may pass.

Feedback and response · Independent

Feedback codes

Staff I ☐ S ☐ C ☐ · Route card feedback boxes: VF ☐ · WS ☐ · I ☐ · E ☐ · NS + … ☐ · R ☐ · // ☐ · ? ☐

Code	Meaning
VF	Verbal feedback
WS	Worked with support
I	Independent
NS + …	Next step
E	Evidence captured
R	Pupil responded to feedback
//	Self-edit point
?	Read this back to me

My response

My response: Record one response to feedback in your usual mode. You may pass.

Adult checks the response and ticks it, or writes R (Pupil responded to feedback).

Made by Matt